MILITARY SPECIFICATION

DETECTOR TICKET, ANTICHOLINESTERASE AGENT, EEL ENZYME

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of detector ticket designed to detect anticholinesterase agents such as GB and VX.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

5-77-2510 - Ticket Packet Assembly
5-77-2528 - Clip

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D 1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Munsell Book of Color

(The Munsell Book of Color may be obtained from the Munsell Color Company, Inc., 2441 N. Calvert Street, Baltimore, MD 21218.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-77-2510 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the detector ticket packet assembly shall conform to Drawing 5-77-2510 and subsidiary drawings.

3.2 Manufacture and assembly. The detector ticket packet shall be manufactured and assembled in accordance with Drawing 5-77-2510 and subsidiary drawings.

3.3 Performance requirements for ambient stored detector tickets.

3.3.1 Cholinesterase recovery before aging. No less than 0.20 and no more than 0.80 cholinesterase units per disc shall be recovered before aging when tested as specified in 4.4.4.1.

3.3.2 VX sensitivity before aging. Detector tickets shall have no color change before aging when tested as specified in 4.4.4.2.

3.3.3 Blank before aging. After 3 minutes, upon challenging with pure water before aging, the enzyme disc (clipped end) shall develop one of the following Munsell colors when tested as specified in 4.4.4.3: hue 7.5BG, 10BG, 2.5B, 5B, 7.5B, 10B; value 7 to 9; chroma 2 or darker.

3.4 Performance requirements for accelerated aging conditions. The detector ticket shall meet the following requirements after accelerated aging for 9 continuous days at 90° ± 2°C.

3.4.1 Cholinesterase recovery after aging. After accelerated aging, no less than 0.20 and no more than 0.80 cholinesterase units per disc shall be recovered when tested as specified in 4.4.4.4.

3.4.2 VX sensitivity after aging. Detector tickets shall have no color change after accelerated aging when tested as specified in 4.4.4.5.

3.4.3 Blank after aging. After 3 minutes, upon challenging with pure water after accelerated aging, the enzyme disc (clipped end) shall develop one of the following Munsell colors when tested as specified in 4.4.4.6: hue 7.5BG, 10BG, 2.5B or 5B; value 7 to 9; chroma 2 or darker.

3.5 pH of water extract. Detector tickets shall have a pH between 8.5 and that of the water before extraction when tested as specified in 4.4.4.7.

3.6 Tear opening. Detector tickets shall be removable from the foil pack without distorting the ticket or using a tool or finger to enlarge the opening when tested as specified in 4.4.4.8.

3.7 Leakage. There shall be no visible bubble leakage any place on the tickets' foil pack when tested as specified in 4.4.4.9.

3.8 Overlay heat seal. The overlay shall strip off so that no more than 1/16 inch of overlay material extends inside the inner edge of the heat seal when tested as specified in 4.4.4.10.

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3.9 Workmanship. The detector tickets and packages shall be free from tears, rips, punctures, cracks, dirt, grease, oil, and poor heat seals.

3.10 Preproduction. Prior to the start of regular production, a preproduction lot of detector tickets shall be produced in accordance with this specification. Regular production techniques, materials, and equipment shall be used to produce the preproduction lot. No more than one lot of each component, reagent, or material shall be used for the preproduction lot.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 525 detector tickets shall be manufactured using the same methods, materials, equipment, and processes of manufacture as will be used during regular production (see 6.2).

4.3.2 Sampling and inspection procedure.

4.3.2.1 For examination. Sampling and inspection procedure for examination shall be as specified in 4.4.2.1 and 4.4.3.1.

4.3.2.2 For test. Sampling shall be conducted in accordance with table I. Tests shall be conducted as specified in 4.4.4.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of tickets shall consist of the detector tickets impregnated with one batch of eel enzyme (see 6.3).

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TABLE I. Preproduction and quality conformance inspection

Requirements	Sample size	Accept/reject criteria	Test paragraph
Cholinesterase recovery before aging	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.1
VX sensitivity before aging	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.2
Blank before aging	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.3
Cholinesterase recovery after aging	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.4
VX sensitivity after aging	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.5
Blank after aging	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.6
pH of water extract	16 tickets per lot	acc 0, rej 1	4.4.4.7
Tear test	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.8
Leakage	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.9
Overlay heat seal	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.10

4.4.2 Sampling.

4.4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling for test shall be conducted in accordance with MIL-STD-105 and table I.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample unit shall be one detector ticket. Sample detector tickets shall be examined in accordance with MIL-STD-105 and the following classification of defects:

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(a) Ticket packet assembly (Drawing 5-77-2510).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Components missing or incorrectly installed
102		Foil pouch sealed area torn, punctured or shows trapped bubbles (this examination must be done prior to assembly of the label)
103		Pouch length or width incorrect
104		Lot number missing
*105		Substrate disc, enzyme disc, or overlay heat seal incorrect
106		Tear notches too deep or missing
107		Workmanship (see 3.9)
Minor:	AQL	2.5 percent defective
201		Pouch seals less than 3/16 inch
202		Label missing or incorrect
203		Position of enzyme or substrate disc incorrect
204		Score line missing

*May be inspected prior to assembly of the ticket into the pouch.

4.4.3.2 For test. Inspection procedure shall be as specified in table I.

4.4.4 Tests (see 6.7). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.4.4.1 Cholinesterase recovery before aging.

(a) pH stat solution. To prepare 1 liter of pH stat solution, dissolve 8.120 grams (g) of magnesium chloride (0.04M), 5.850 g of sodium chloride (0.10M), and 0.730 g of acetylcholine chloride or 0.909 g of acetylcholine bromide in 900 milliliters (mL) of deionized water having a pH between 5.0 and 6.0 (the pH of final solution will be approximately 4.5). Adjust the final volume to 1 liter with deionized water. (Solution is stable for 10 consecutive calendar days if kept refrigerated at 5° ± 5°C.)

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(b) Procedure. To a 100-mL beaker add 50 mL of pH stat solution prepared as specified in (a). Equilibrate this solution in a water bath to 37° ± 1°C. To a 20-mL beaker add 2.0 mL of deionized water having a pH between 7.0 and 7.2. Peel off one eel acetylcholinesterase enzyme-impregnated paper disc from the clipped end of the ticket and add to the beaker. (Scrape off as much of disc as possible from around seal area and also add these scrapings to the beaker.) Break up the paper disc with a glass rod and then stir vigorously at room temperature for 5 minutes with a magnetic stirrer. While this is mixing, adjust the pH of the pH stat solution to 8.0 using an automatic titrator (pH stat type), using 0.004N NaOH and maintaining 37° ± 1°C. After temperature and pH have equilibrated, let the solution stir an additional 5 minutes while maintaining a pH of 8.0. Record the volume of NaOH added as the blank. Pour approximately 15 mL of pH 8.0 pH stat solution into the 20-mL beaker with enzyme disc and then quickly transfer the mixture back into the 100-mL titration beaker which is placed back into the 37°C bath. Start the timer immediately upon mixing. Record the buret reading and start titration. The automatic titrator will titrate to and control the pH at 8.0. While the titration is running, record the buret reading at 2 minutes. After 7 minutes, stop the titration and record the final buret reading. Calculate the cholinesterase units per disc recoverd as follows:

$$\text{Recovered units per disc} = \frac{1000\ (A - D)\ B}{C}$$

where: A = Milliliters of NaOH used between 2 and 7 minutes,
B = Normality of NaOH,
C = Time in minutes, and
D = Milliliters of NaOH used during the 5 minute blank period.

4.4.4.2 VX sensitivity before aging.

(a) Preparation of VX agent stock solutions. Dilute a quantity of VX of known purity, based on an assay performed within the last 60 days (see 6.6 for a suggested method), to 100 mL in a volumetric flask with isopropyl alcohol so that the solution contains 10 microliters of pure VX (Stock Solution A). Dilute 0.1 mL of Stock Solution A to 100 mL in a volumetric flask with deionized water having a pH of between 5.0 and 6.0 (Stock Solution B). Dilute 2 mL of Stock Solution B to 10 mL with deionized water in a volumetric flask to produce a working solution of 0.02 milligrams (mg) VX per liter. Stock solutions shall not be more than seven consecutive calendar days old when used. The pH of the deionized water may be adjusted with hydrochloric or sulfuric acid if necessary.

(b) Ticket agent VX testing procedure. Wet a new enzyme disc (clipped end) by dipping it in 65° to 75°F VX working solution for 5 seconds. If the enzyme disc does not completely wet, replace the foil overlay and lightly press the area over the enzyme disc to saturate the entire surface. More VX working solution can be added if necessary. After the enzyme disc is completely wet, fold the silver cover back over the patch and fold the ticket.

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Insert in clip (Drawing 5-77-2528) and hold in fist for 3 minutes. Peel off the foil overlay to expose the disc on the square corner end. Fold the ticket along the crease so that the two discs contact each other. Press together, insert in clip, and hold in hand for 3 minutes. After reacting, open the ticket and observe the color of the enzyme disc (clipped corner end). The disc must remain white to pass the test.

4.4.4.3 Blank before aging (ticket inhibition). Wet the enzyme disc (clipped end) by dipping it in 65° to 75°F deionized water for 5 seconds. If the enzyme disc does not completley wet, replace the foil overlay and lightly press the area over the enzyme disc to saturate the entire surface. More water can be added if necessary. After the enzyme disc is completely wet, insert in clip and hold in hand for 3 minutes. Peel off the foil overlay to expose the substrate disc (square corner end). Fold ticket along crease so that the two paper discs contact each other. Press together, insert in clip, and hold in hand for 3 minutes. After reacting, open the ticket and observe the color of the enzyme disc (clipped corner end). The paper disc must turn blue to pass the test.

4.4.4.4 Cholinesterase recovery after aging. Place the required amount of complete ticket assemblies in an air circulating oven at the temperature and for the time specified in 3.4. Remove the tickets from the oven and store them at room temperature (65° to 75°F) for at least 1 hour. Perform the cholinesterase recovery test specified in 4.4.4.1.

4.4.4.5 VX sensitivity after aging. Place the required number of complete ticket assemblies in an air circulating oven at the temperature and for time specified in 3.4. Remove the tickets from the oven and then store at room temperature (65° to 75°F) for at least 1 hour. Perform the VX sensitivity test specified in 4.4.4.2. The disc must remain white to pass the test.

4.4.4.6 Blank after aging. Place the required number of complete ticket assemblies in an air circulating oven at the temperature and for the time specified in 3.4. Remove the tickets from the oven and then store at room temperature (65° to 75°F) for at least 1 hour. Perform the pure water challenge test specified in 4.4.4.3. The disc must turn blue to pass the test.

4.4.4.7 pH of water extract. Remove 16 enzyme discs from the detector tickets and place them in a 30-mL beaker containing 5 mL of water at a pH between 6.0 and 7.0. Break up the discs with a glass rod and agitate the mixture vigorously for 5 minutes. Determine the pH using a pH meter.

4.4.4.8 Tear test. The tear test samples must have been stored at room temperature (65° to 75°F) for at least 1 hour prior to tearing. All external moisture on the package should be dried prior to tearing. With bare hands grasp the ticket on each side of the tear notch and tear the ticket towards the opposite notch with a snap of the wrists. The tear must satisfy the conditions of 3.6. If not, a second attempt is permitted by tearing from the opposite tear notch.

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4.4.4.9 Leakage. This test must be done prior to application of the label to the pouch. Place the sample tickets in a clear vacuum chamber and hold them vertically by the long heat seal opposite the tear notches. Fill with tap water at 65° to 75°F to 1 to 1-1/2 inches above the highest ticket's edge or corner. Close the chamber and evacuate to an absolute pressure above the water surface of 310 ± 30 millimeters (mm) Hg. Continuously examine the packets during evacuation and for 60 seconds after establishing 310 mm Hg. No bubble leakage is permitted any place on the surface of the pouch or pouch heat seals.

4.4.4.10 Overlay heat seal. Peel back overlay to expose substrate disc and examine for the requirements of 3.8.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The detector ticket covered by this specification is intended for use in chemical agent detector kits for the detection of anti-cholinesterase chemical agents in water.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification and preproduction instructions.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of E29.

6.5 Manufacture and assembly. The following methods are furnished to the contractor with the distinct understanding that (a) the contractor is not required to follow the methods, (b) the requirements of the specification must be met regardless of the methods of manufacture, and (c) the Government makes no warranty that material manufactured in accordance with these methods will meet the specification requirements.

6.5.1 Preparation of eel enzyme medium. Prepare a solution of piperazine-N,N' bis (2-hydroxypropane sulfonic acid)·2 H_2O (POPSO), 0.5M, by suspending the solid buffer in 30 mL of distilled water. Adjust the pH to 8.0 by adding 5M sodium hydroxide. Add 0.2 g of Bovine Serum Albumin (Fraction V, Sigma Chemical Company or equal) and 0.6 mL of 1.0 percent "Triton" X-100 solution. Dilute to 40 mL with distilled water.

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6.5.2 Preparation of eel enzyme solution. Dissolve a known quantity of eel cholinesterase in the medium described in 6.5.1 such that the resulting solution contains 200 units per mL. Assay this solution in accordance with the procedure given in 4.4.4.1 using 0.01 mL of the solution instead of the enzyme disk. Based on this assay, adjust the enzyme concentration to 36 units per mL using the medium.

6.5.3 Preparation of eel enzyme discs. Lay discs of Whatman DE81 Ion Exchange Paper, 1/2 inch diameter, on thoroughly cleaned glass plates. Pipet 0.03 ± 0.005 mL of enzyme solution onto each ticket with a calibrated pipet or syringe. Allow the discs to air dry for 1 to 3 hours, then vacuum dry for at least 16 hours at an absolute pressure of 0.1 mm Hg or less. Store over desiccant in a refrigerator until needed. NOTE: Enzyme impregnated discs should always be protected from moisture.

6.5.4 Preparation of oxidizer solution. In 100 mL of deionized water, pH 5.0 to 6.0, dissolve 0.0448 g potassium ferricyanide ($K_3Fe(CN)_6$) and 0.0715 g potassium ferrocyanide ($K_4Fe(CN)_6 \cdot 3H_2O$).

6.5.5 Preparation of indoxyl acetate solution. Prepare an 18 mg per mL solution of indoxyl acetate in acetone immediately before use and prepare only the amount needed. Protect from light.

6.5.6 Preparation of indoxyl acetate substrate discs. Arrange 9.0 centimeter (cm) circles of number 1 filter paper (Whatman, Inc.) on a clean glass plate so that none touch. Pipet 1.0 mL oxidizer solution (see 6.5.4) onto each circle. Insure that the entire circle is wetted with the solution. Allow to air dry 2 to 3 hours. Reduce lighting to a minimum to protect the indoxyl acetate, and then pipet 1.0 mL of indoxyl acetate solution onto each circle. Insure that the entire circle is wetted with the solution. Allow to dry for 30 minutes. Cut into 5/8 inch diameter discs. Store over desiccant protected from light and under refrigeration until needed. NOTE: Indoxyl acetate in solution and impregnated discs should always be protected from moisture and light.

6.5.7 Assembly and packaging of detector ticket. The assembly steps should be done in a room where the temperature does not exceed 75°F and the relative humidity does not exceed 50 percent. Before assembling the ticket, wash the ticket backsheet (Drawing 5-77-2515) in a hot Alconox solution in an ultrasonic cleaning bath; rinse in deionized water and air dry. Degas these in a desiccator at 0.01 mm Hg or less for 13 to 24 hours. Then assemble the ticket as described on Drawing 5-77-2514. Place the completed ticket into the ticket packet (Drawing 5-77-2511) and insert the impregnated pad (Drawing 5-77-2513) which has been freshly activated by heating to 150°C for 2 hours. Place the unsealed ticket packets in a vacuum desiccator. Vacuum dry for 16 to 24 hours at 0.01 mm Hg or less at 25°C. Remove the ticket packets and heat seal immediately so that exposure to atmospheric humidity is kept to a minimum. Assembled tickets not in packets and assembled packets which are not sealed should not be left undesiccated for more than 1 hour. If overnight storage is required, they should be desiccated under vacuum.

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6.6 Optional procedure for determining purity of VX. From a VX standard of known purity greater than 80 percent as determined by volumetric methods, prepare a solution containing 5 to 10 mg per mL of VX and an equal weight of methyl laurate in isopropanol. Assay using a gas chromatograph equipped with a flame ionization detector and a means of integrating peak areas.

(a) Reagents. Isopropanol, reagent grade; methyl laurate, 99 percent pure for internal standard grade; VX standard, 80 percent or greater purity; chromatographic gases: nitrogen 99.995 percent, hydrogen 99.95 percent; and compressed air, oil free, best grade available.

(b) Equipment. Gas chromatograph, temperature programmed with flame ionization detector using the above gases. Each gas cylinder should be supplied with a two stage regulator to reduce pressure to 0 to 100 pounds per square inch gage. The hydrogen and helium lines should have filters to remove particulates, moisture, hydrocarbons, and oxygen. Chromatographic parameters are shown in table II.

(c) Calibration curve. Determine the peak area of the methyl laurate and VX standard peaks for several concentrations of VX. The calibration points are obtained from the following formula:

$$K = \frac{AD}{BC}$$

where: A = Weight of VX standard, in grams,
B = Weight of methyl laurate internal standard, in grams,
C = Area of VX peak, and
D = Area of methyl laurate peak.

From several K values, a calibration curve can be drawn.

TABLE II. Chromatographic parameters

Characteristic	Parameter
Column material	Glass, 2 mm ID by 6 feet long
Column packing	4 percent SE-30 and 6 percent OV-10 on 80-100 mesh "Gas Chrom Q"
Injection port temperature	225°C
Detector temperature	250°C
Column temperature	Start at 160°C, hold for 1 minute. Increase at a programmed rate of 10°C per minute to 225°C, hold for 5 minutes.
Carrier gas flow	45 cc per minute (N_2)
Hydrogen flow	40 cc per minute
Air flow (for flame)	300 cc per minute

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(d) Calculation. For the unknown VX, weigh a sample into isopropanol, add the known weight of internal standard and inject into the chromatograph. Determine the percent by weight VX as follows:

$$\text{Percent VX} = \frac{KEG}{FH}$$

where: E = Weight of standard, in grams,
F = Weight of VX, in grams,
G = Area of VX peak,
H = Area of methyl laurate peak, and
K = Value from calibration curve prepared in 6.6(c).

6.7 Testing precautions. This specification covers the use of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials. This specification requires testing with Chemical Surety Material (CSM) which is potentially hazardous to personnel. Testing with CSM shall be conducted in a facility which has been approved for this type of operation by the US Army. All general and specialized procedures which include handling, using, and storing CSM must be in compliance with the contract clause "Safety Provisions for Contracts Involving Chemical Surety Material and Other Related Military Unique Chemical Compounds", dated 1 April 1984. Additionally, this specification requires the use of a known carcinogen.

6.8 Supersession data. This specification supersedes and includes the requirements of Chemical Systems Laboratory Purchase Description EA-D-1296, dated 3 January 1983.

Custodian:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-51503(EA)	DETECTOR TICKET, ANTICHOLINESTERASE AGENT, EEL ENZYME

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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